a distance of $l/8$. Initially, C's x was $l/2$. After t, C still misses the y axis, so it cannot collide with D.

2.

Launch gives the craft enough kinetic energy to escape Earth with the speed needed for its heliocentric ellipse. Far from Earth, its solar distance still equals the Earth-Sun distance. It enters the ellipse from Earth's orbit at E (Fig. 1). The circle of radius r_{se}, A is Earth's orbit, with the Sun S at its center. Ellipse B is the craft's solar orbit, and P is its perihelion.

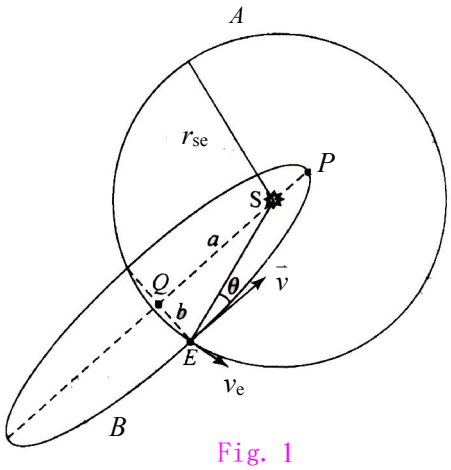


Fig. 1

The craft's period equals Earth's. By Kepler's third law, the ellipse's semimajor axis a equals the Earth-Sun distance r_{se}. Thus

$$a = r_{se} \qquad (1)$$

Distances to an ellipse's two foci sum to $2a$; thus the orbits' intersection point E is a minor-axis vertex; E and Q (the ellipse center) define a line normal to the major axis. Triangle $\triangle ESQ$ gives the semiminor axis

$$b = \sqrt{r_{se}^2 - (a - \overline{SP})^2} \ . \qquad (2)$$

From (1), (2) and the values $a = r_{se} = 1\text{AU}$, $\overline{SP} = 0.01\ \text{AU}$ we obtain

$$b = 0.141\text{AU} \ . \qquad (3)$$

In the Sun's frame, the craft's angular momentum and mechanical energy are conserved. Set the speed at E to v, the perihelion speed to v_p, the mass to m, the solar mass to M_s. Then

$$mva\sin\theta = mv_p\overline{SP} \ , \qquad (4)$$

Here θ is the angle between velocity v and the line joining E, S:

$$\sin\theta = \frac{b}{a} \ . \qquad (5)$$

Conservation of mechanical energy gives